Reproduction quickstart

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Everything needed to re-run the results in the mentor-review draft, in the order a reviewer would want them.

1. Get the source

```
git clone https://github.com/uskutsav-cpu/selfdual-5form-invariants
cd selfdual-5form-invariants
git switch publication/jhep-mentor-draft
```

2. Environment

Python 3.10 or newer is required. This is not optional and it is not documented anywhere else in the repository — it was found by a clean-clone run. The source uses PEP 604 union annotations (`X | Y`), so on Python 3.9 both test suites fail at *collection* with `TypeError: unsupported operand type(s) for |` before a single test runs. On a stock macOS, `python3` is 3.9.6 and the obvious command below will therefore fail unless you name a newer interpreter. The certified results were produced under Python 3.13.

```
python3.13 -m venv .venv        # or any python >= 3.10
.venv/bin/python3 -m pip install -r requirements.txt
```

Dependencies are NumPy, `pynauty` and `pytest`.

To rebuild the figures and the manuscript you additionally need:

```
.venv/bin/python3 -m pip install -r manuscript/requirements-docs.txt
```

which installs matplotlib, plus a TeX engine (Tectonic 0.17.0 was used) and, only for the mentor-package documents, pandoc. These are kept out of the top-level `requirements.txt` deliberately: that file pins the environment the science was certified in, and nothing in the docs set participates in a computation whose result is quoted.

Two notes that will otherwise cost you time:

- `opt_einsum` is **required** and is now in `requirements.txt`. This corrects an earlier note here that called it optional and said the certified runs ran without it. It supplies the contraction order *and* the intermediate-size and flop estimates that the modular contractor enforces as a budget. Without it the old code took an unguarded naive contraction order: on an 8 GiB host `StructuredDegree8.values` goes from 0.4 s at 0.27 GB peak to an unbounded allocation that the kernel kills, and the bridge suite dies with SIGKILL and no traceback. The fallback now raises instead of running unguarded. Values are unchanged — the 86 bridge tests pass identically once it is installed.
- If you use the repository's committed `.venv`, its `pip` shebang points at a path that no longer exists. Use `.venv/bin/python3 -m pip` rather than `.venv/bin/pip`.

3. Test suites

```
.venv/bin/python3 -m pytest tests -q
.venv/bin/python3 -m pytest spinor_trace_bridge/tests -q
```

Expected: **254** tensor tests and **86** bridge tests, no failures.

4. The exact rational certificates

```
.venv/bin/python3 scripts/d10_characteristic_zero.py
.venv/bin/python3 scripts/b10_p10_characteristic_zero.py
```

Expected:

quantity	value
`dim_Q A10`	14
`dim_Q D10`	11
`dim_Q Q10`	3
`dim_Q(B10   P10)`	1

The first script lifts 9 non-integral rows by CRT across 5 primes, validates at held-out prime 32771, and runs the closure to a fixed point in `Fraction` arithmetic (3 sweeps). The second lifts all 12 published structures across 7 primes, validates at held-out prime 32783, and re-verifies the generator identity at fresh prime 32869 on 6 fresh samples.

5. The rank matrix

```
.venv/bin/python3 scripts/aggregate_rank_matrix.py --self-test
```

Read-only with respect to the cells and order-independent, so it is safe to run at any time. Expected: 15 cells, rank 81 in every one, 81 stable pivot rows and 81 stable pivot columns, 0 unstable.

Caution. Individual cells may be run in parallel but must not share a row cache. Two processes writing one cache is a failure this project has actually experienced.

6. Rebuild the manuscript

```
.venv/bin/python3 manuscript/scripts/make_numbers.py
.venv/bin/python3 manuscript/jhep/scripts/make_figures.py
.venv/bin/python3 manuscript/jhep/scripts/make_tables.py
.venv/bin/python3 manuscript/jhep/scripts/check_draft.py
cd manuscript/jhep && tectonic -X compile main.tex
```

The macro, figure and table steps must precede the compile: the manuscript contains no scientific value of its own, only macros that expand to values read from artifacts.

`check_draft.py` runs 52 gates covering LaTeX health, citation coverage, placeholder discipline, and the wording restrictions (no unscoped completeness or canonicality claims, rank 81 scoped to the selected family, degree-eight claim qualified, degree-twelve scope stated).

Expected: **52 gates passed, 0 failed**, and a 41-page PDF with zero overfull boxes, zero underfull boxes, zero undefined citations and zero undefined references.

7. A note on TeX

The draft compiles with **Tectonic**. Because `jheppub.sty` requests `hyperref` with `pdfa=true` — which selects the pdfTeX driver and fails under Tectonic's XeTeX engine — `main.tex` passes `xetex` to `hyperref` before the class is loaded. The official JHEP style file is not modified. Compiling with pdfLaTeX instead should also work and does not need that line.

8. Determinism, and its limit

Figures are byte-identical across repeated runs **in a fixed environment**; matplotlib's PDF creation timestamp is suppressed explicitly. Archives use a fixed epoch and normalised member metadata.

Byte-identity does **not** hold across matplotlib versions. A clean-clone run that resolved matplotlib 3.9.4 produced figures visually identical to, and byte-different from, those built with 3.11.1. That is why `manuscript/requirements-docs.txt` pins the version exactly. If you install the pinned version and the figure hashes still differ, that is a finding. If you install a different version, expect different bytes and identical pictures.